OAK RIDGE National Laboratory

AI-Agent Driven Survey and Validated Literature Review for Discovering Solid Breeder Materials

End-to-End Traceability for Automated Literature Generation

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Denver, Colorado

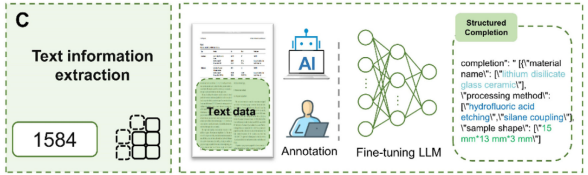
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U.S. DEPARTMENT OF ENERGY

Good afternoon. I'm Jose Tupayachi, from ORNL. Today I'm presenting our end-to-end framework for AI-agent driven literature generation .

AI-Agent Driven Literature Reviews with Full Claim Traceability



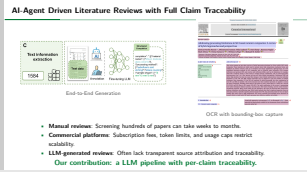
End-to-End Generation



OCR with bounding-box capture

- **Manual reviews**: Screening hundreds of papers can take weeks to months.
- **Commercial platforms**: Subscription fees, token limits, and usage caps restrict scalability.
- **LLM-generated reviews**: Often lack transparent source attribution and traceability.

Our contribution: a LLM pipeline with per-claim traceability.



Explain

Solid Breeders are one the Bottleneck for Fusion Energy

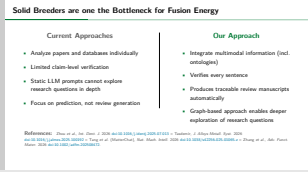
Current Approaches

- Analyze papers and databases individually
- Limited claim-level verification
- Static LLM prompts cannot explore research questions in depth
- Focus on prediction, not review generation

Our Approach

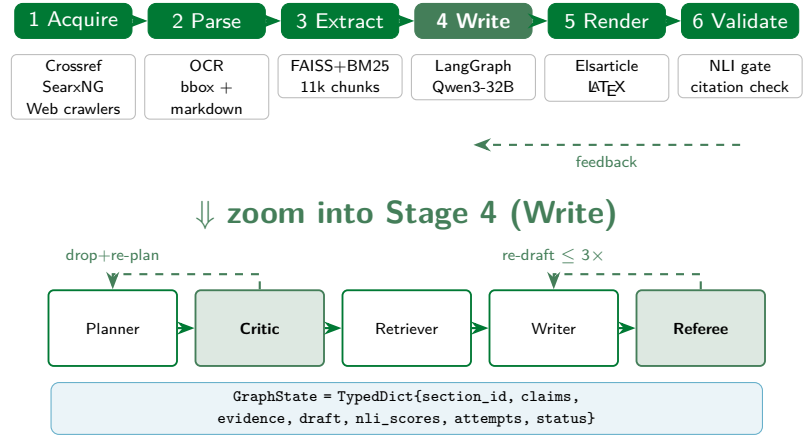
- Integrate multimodal information (incl. ontologies)
- Verifies every sentence
- Produces traceable review manuscripts automatically
- Graph-based approach enables deeper exploration of research questions

References: Zhou *et al.*, *Int. Dent. J.* 2026 doi:10.1016/j.identj.2025.07.013 ○ Tasdemir, *J. Alloys Metall. Syst.* 2026 doi:10.1016/j.jalmes.2025.100192 ○ Tang *et al.* (MatterChat), *Nat. Mach. Intell.* 2026 doi:10.1038/s42256-025-01065-z ○ Zhang *et al.*, *Adv. Funct. Mater.* 2026 doi:10.1002/adfm.202508472.

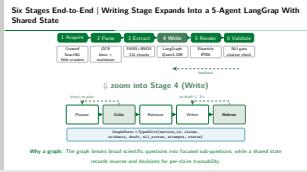


Recent LLM work in materials science typically addresses individual research tasks through single prompts or reasoning chains. While effective for information extraction and question answering, these approaches do not produce complete, citation-traceable review manuscripts. Our framework integrates these capabilities into a validated end-to-end system.

Six Stages End-to-End | Writing Stage Expands Into a 5-Agent LangGrap With Shared State



Why a graph: The graph breaks broad scientific questions into focused sub-questions, while a shared state records sources and decisions for per-claim traceability.

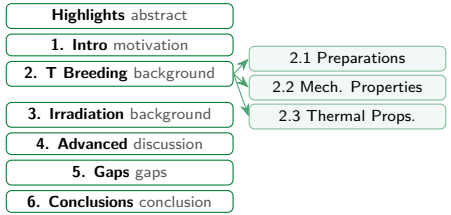


The pipeline starts from a single reference paper. It produces two intermediate files that remain editable: one captures the structure—sections and research questions—and the other captures style, including voice, tense, and rhetorical patterns.

Ontologies Tell Agents What and How to Write

Structure ontology: What to write

18 sections extracted from the reference paper:



Each node → {role, purpose, expected_evidence, research_questions}.

Style ontology: How to write

```
hedging: medium
citation: numeric
```

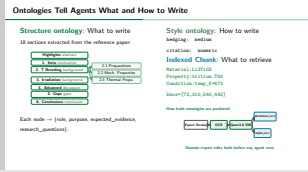
Indexed Chunk: What to retrieve

```
Material:Li2TiO3
Property:tritium.T50
Condition:temp_K=673
bbox=[72,310,540,442]
```

How both ontologies are produced:



Domain expert edits both before any agent runs.

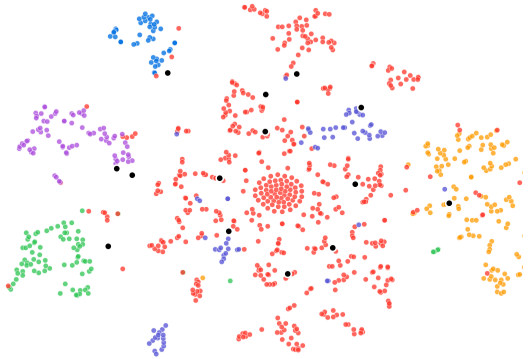


These ontologies act as shared contracts between agents. For example, in selecting lithium solid breeder materials for fusion systems, agents rely on both procedural knowledge (how to navigate design and selection steps) and semantic knowledge (how to interpret material options and properties), both of which can be formalized as ontologies over entities, relations, and constraints.

Critic and Referee Coverage are Structured Before Writing

Corpus embedding

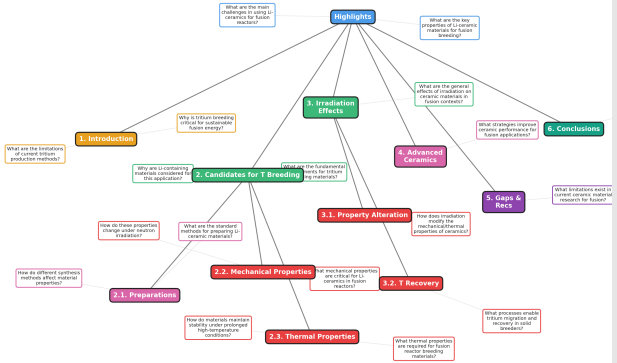
984 chunks · 14 papers · TF-IDF + t-SNE, k-means (k=6)



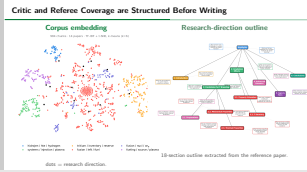
hidrojen / hte / hydrogen
systems / injection / plasma
tritium / inventory / reserve
fusion / left / fuel
fusion / nucl / org
fueling / source / plasma

dots = research direction.

Research-direction outline



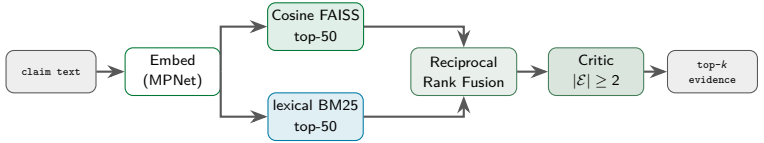
18-section outline extracted from the reference paper.



The Critic rejects claims without at least two supporting chunks. The Referee filters sentences below 0.55 entailment or 10% numeric error. A final Q score combines grounding, citations, structure, and style.

Per-Claim Paper Selection: Hybrid Lexical and Semantic Retrieval: Evidence Search

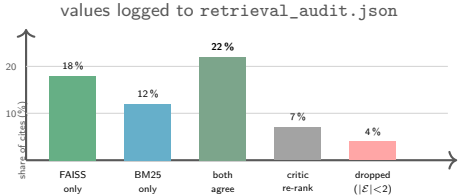
Critic (pre-write): $|\mathcal{E}_i| \geq 2$ with distinct sources, else drop. **Referee** (post-write): $P_{\text{ent}} \geq 0.55 \wedge |\hat{y} - y|/|y| \leq 0.10$, else re-draft. **Composite:** $Q = 0.3\,S_g + 0.3\,V_{\text{cite}} + 0.2\,S_f + 0.2\,S_{\text{sty}}$.

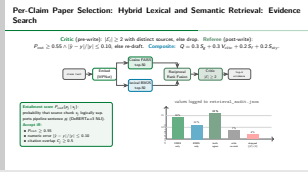


Entailment score $P_{\text{ent}}(p_j \mid s_j)$:
probability that source chunk s_j logically supports pipeline sentence p_j (DeBERTa-v3 NLI).

Accept iff:
- $P_{\text{ent}} \geq 0.55$
- numeric error $|\hat{y} - y|/|y| \leq 0.10$
- citation overlap $C_j \geq 0.5$

values logged to `retrieval_audit.json`



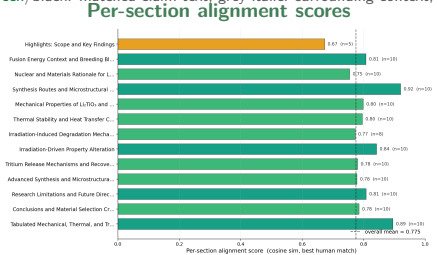


Our hybrid approach uses two retrieval methods: BM25, which finds exact keyword matches, and FAISS, which finds semantically similar content.
Running both in parallel improves coverage, since each can capture evidence the other may miss. When both point to the same chunks, it increases confidence in the final claim.

Per-Claim Traceability in Action: Accepted Claims are Incorporated. Failed Claims Are Dropped (Referee)

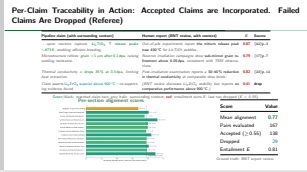
Pipeline claim (with surrounding context)	Human expert (BNT review, with context)	E	Source
. . . upon neutron capture, Li_2TiO_3 T release peaks ∼673 K, *enabling efficient breeding.*	*Out-of-pile experiments report* the tritium release peak near 400 °C *for* Li_2TiO_3 *pebbles.*	0.87	[42]p.3
Microstructure refines: grain <1 μm after 0.1 dpa, *raising swelling resistance.*	*Neutron irradiation campaigns show* sub-micron grain refinement above 0.05 dpa, *consistent with TEM observations.*	0.79	[17]p.7
Thermal conductivity κ drops 35 % at 0.3 dpa, *limiting heat extraction.*	*Post-irradiation examination reports a* 30–40 % reduction in thermal conductivity *at comparable dose levels.*	0.82	[23]p.12
Claim asserts Li_2ZrO_3 superior above 900 °C – *no supporting evidence found.*	*(BNT review discusses* Li_2ZrO_3 *stability but reports* no comparative performance above 900 °C.*)*	0.41	drop

Green/black: matched claim text; grey italic: surrounding context; **red**: entailment score E; last row dropped ($E < 0.55$).



Score	Value
Mean alignment	**0.77**
Pairs evaluated	167
Accepted (≥ 0.55)	138
Dropped	29
Entailment E	0.81

Ground truth: BNT expert review.



Accepted sentences pass entailment, numeric accuracy, and citation checks, then appear in the manuscript with clickable links to their exact source (paper ID, page, and location). If a sentence fails any check, it is removed entirely. For example, a claim about lithium zirconate above 900°C was dropped because its entailment score (0.41) was too low.

Side-by-Side Comparison: Human and Pipeline Manuscripts Agree on the Same Claims

Human-expert manuscript Expert review, p. 1

this element in nature. Production of tritium is achieved by neutron irradiation of lithium (Li) containing materials. Prominent tritium producing materials, known as breeders, are classified as either solids or liquids. The review here focuses on solid materials such as Li_2TiO_3, Li_2ZrO_3, Li_4SiO_4, and advanced

Performance of irradiation campaigns using neutron, ion beam, and other sources are explored as well as exploration of the literature for advanced ceramics such as biphasic or 3D printed materials. Utilization of

influence on grain structure, density, porosity, and phase stability. Examination of literature offers insight into the mechanical and thermal properties of solid breeders and the irradiation induced alterations.

Pipeline-generated manuscript `paper.pdf`, p. 10–11

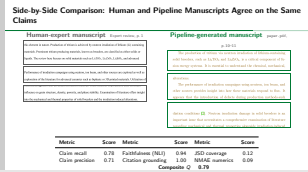
The production of tritium via neutron irradiation of lithium-containing solid breeders, such as Li_2TiO_3 and Li_2ZrO_3, is a critical component of fusion energy systems. It is essential to understand the chemical, mechanical,

alterations.

The performance of irradiation campaigns using neutron, ion beam, and other sources provides insight into how these materials respond to flux. It appears that the introduction of defects during production methods-such

diation conditions [3]. Neutron irradiation damage in solid breeders is an important issue that necessitates a comprehensive examination of literature regarding mechanical and thermal properties alongside irradiation-induced

Metric	Score	Metric	Score	Metric	Score
Claim recall	0.78	Faithfulness (NLI)	0.94	JSD coverage	0.12
Claim precision	0.71	Citation grounding	1.00	NMAE numerics	0.09
				Composite Q	**0.79**

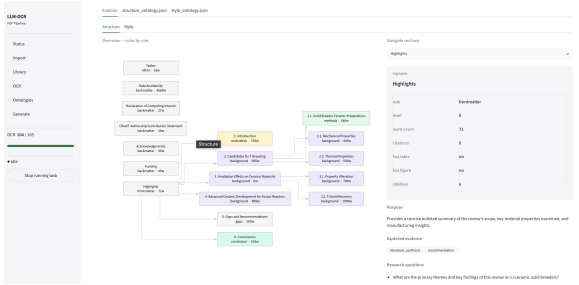


The Full Stack Runs Locally on One GPU - One Click Deployment



Docker containers orchestrate the stack.
Ollama serves local LLM inference.

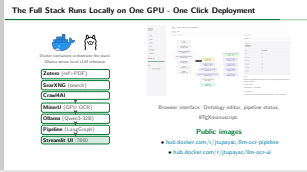
- **Zotero** (ref+PDF)
- **SearXNG** (search)
- **Crawl4AI**
- **MinerU** (GPU OCR)
- **Ollama** (Qwen3-32B)
- **Pipeline** (LangGraph)
- **Streamlit UI** :7860



Browser interface: Ontology editor, pipeline status, LaTeX manuscript.

Public images

- hub.docker.com/r/jtupayac/llm-ocr-pipeline
- hub.docker.com/r/jtupayac/llm-ocr-ui



The whole system runs fully on a single 48GB GPU, The system is sandboxed so no data ever leaves the machine.
Setup is quick: just two Docker pulls and one command. It launches all the neccessary tools and packages.
Now I'll open the live interface.
[CLICK — wait 20 seconds]
The interface has three parts: a top outline tree where you can edit nodes like roles, questions, and evidence; a middle panel showing the live agent workflow; and a bottom PDF preview that updates in real time. Everything runs in one browser tab—no coding needed.

Questions?

Here is a side-by-side. The human expert writes a sentence on irradiation-driven property alteration. The pipeline independently generates the same claim, anchored to a specific bounding box in the source manuscript. We show an alignment score: 0.86.

Backup: Validation Metrics

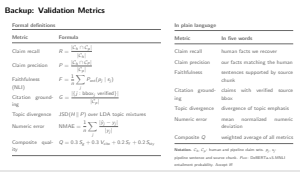
Formal definitions

Metric	Formula
Claim recall	$R = \dfrac{\lvert\mathcal{C}_h \cap \mathcal{C}_p\rvert}{\lvert\mathcal{C}_h\rvert}$
Claim precision	$P = \dfrac{\lvert\mathcal{C}_h \cap \mathcal{C}_p\rvert}{\lvert\mathcal{C}_p\rvert}$
Faithfulness (NLI)	$F = \dfrac{1}{n}\sum_j P_{\text{ent}}(p_j \mid s_j)$
Citation grounding	$G = \dfrac{\lvert\{j : \text{bbox}_j \text{ verified}\}\rvert}{\lvert\mathcal{C}_p\rvert}$
Topic divergence	$\text{JSD}(H \,\Vert\, P)$ over LDA topic mixtures
Numeric error	$\text{NMAE} = \dfrac{1}{n}\sum_j \dfrac{\lvert\hat{y}_j - y_j\rvert}{\lvert y_j\rvert}$
Composite quality	$Q = 0.3\,S_g + 0.3\,V_{\text{cite}} + 0.2\,S_f + 0.2\,S_{\text{sty}}$

In plain language

Metric	In five words
Claim recall	human facts we recover
Claim precision	our facts matching the human
Faithfulness	sentences supported by source chunk
Citation grounding	claims with verified source bbox
Topic divergence	divergence of topic emphasis
Numeric error	mean normalized numeric deviation
Composite Q	weighted average of all metrics

Notation. $\mathcal{C}_h$, $\mathcal{C}_p$: human and pipeline claim sets. p_j, s_j: pipeline sentence and source chunk. P_{ent}: DeBERTa-v3-MNLI entailment probability. Accept iff



Additional Definitions

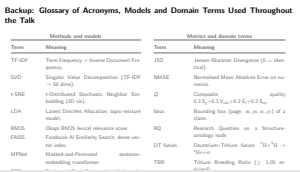
Backup: Glossary of Acronyms, Models and Domain Terms Used Throughout the Talk

Methods and models

Term	Meaning
TF-IDF	Term Frequency × Inverse Document Frequency.
SVD	Singular Value Decomposition (TF-IDF → 50 dims).
t-SNE	t-Distributed Stochastic Neighbor Embedding (2D vis).
LDA	Latent Dirichlet Allocation; topic-mixture model.
BM25	Okapi BM25 lexical relevance score.
FAISS	Facebook AI Similarity Search; dense vector index.
MPNet	Masked-and-Permuted sentence-embedding transformer.

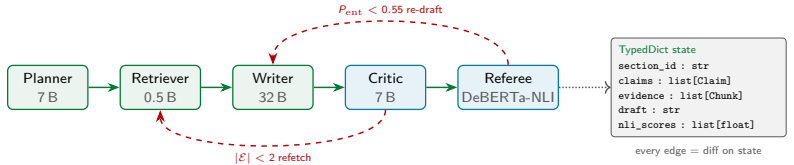
Metrics and domain terms

Term	Meaning
JSD	Jensen–Shannon Divergence (0 = identical).
NMAE	Normalised Mean Absolute Error on numerics.
Q	Composite quality $0.3\,S_g{+}0.3\,V_{\text{cite}}{+}0.2\,S_f{+}0.2\,S_{\text{sty}}$.
bbox	Bounding box (page, x_0, y_0, x_1, y_1) of a claim.
RQ	Research Question on a Structure-ontology node.
DT fusion	Deuterium–Tritium fusion: ${}^2\text{H}{+}{}^3\text{H} \rightarrow {}^4\text{He}{+}n$.
TBR	Tritium Breeding Ratio ($\geq$ 1.05 required)

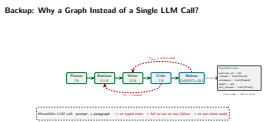


TF-IDF — Term Frequency × Inverse Document Frequency A classic way to score how "important" a word is to a document. Term Frequency counts how often a word appears in a doc; Inverse Document Frequency penalizes words that appear in almost every document (like "the"). The product tells you which words are distinctive to a given document rather than just common everywhere. SVD — Singular Value Decomposition A linear-algebra factorization that decomposes a matrix into three simpler matrices. Here it's applied to a TF-IDF matrix to compress documents into 50 numbers (dimensions) that capture latent meaning — this is also called Latent Semantic Analysis (LSA). Similar documents end up with similar 50-dim vectors. t-SNE — t-Distributed Stochastic Neighbor Embedding A dimensionality-reduction algorithm used purely for visualization. It takes high-dimensional vectors (like those 50-dim SVD embeddings) and squashes them down to 2D while trying to preserve which points are neighbors. The result is a scatter plot where clusters of similar documents appear as visible blobs. LDA — Latent Dirichlet Allocation A probabilistic topic model. It assumes each document is a mixture of topics, and each topic is a distribution over words. You tell it "find K topics," and it learns both what words define

Backup: Why a Graph Instead of a Single LLM Call?



Monolithic LLM call: prompt → paragraph × no typed state × full re-run on any failure × no per-claim audit



We may be wondering why build this as a graph rather than a Chain of Though call? Because a graph gives us fine-grained control over the workflow. Each step can be executed independently, and conditional loops allow revisions flows. The workflow auditable, enabling every claim to be traced back to its supporting evidence.

Backup: Validation Protocol: Every Sentence Is Scored Against a Human Expert

Alignment protocol

1. Split human & pipeline drafts into sentences.
2. For each p_j find nearest $h_{\pi(j)}$ in MPNet space.
3. Score the pair on three axes:

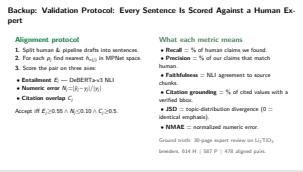
- **Entailment** E_j — DeBERTa-v3 NLI
- **Numeric error** $N_j = |\hat{y}_j - y_j| / |y_j|$
- **Citation overlap** C_j

Accept iff $E_j \geq 0.55 \wedge N_j \leq 0.10 \wedge C_j \geq 0.5$.

What each metric means

- **Recall** = % of human claims we found.
- **Precision** = % of our claims that match human.
- **Faithfulness** = NLI agreement to source chunks.
- **Citation grounding** = % of cited values with a verified bbox.
- **JSD** = topic-distribution divergence (0 = identical emphasis).
- **NMAE** = normalized numeric error.

Ground truth: 30-page expert review on Li_2TiO_3 breeders. 614 H | 587 P | 478 aligned pairs.



We verify quality against a 30-page expert review on lithium titanate breeders (614 human sentences vs. 587 generated). Each generated sentence is matched to the closest human sentence in embedding space, then checked on three criteria: NLI entailment (DeBERTa), numeric accuracy, and citation overlap. A sentence is accepted only if it passes all three thresholds.